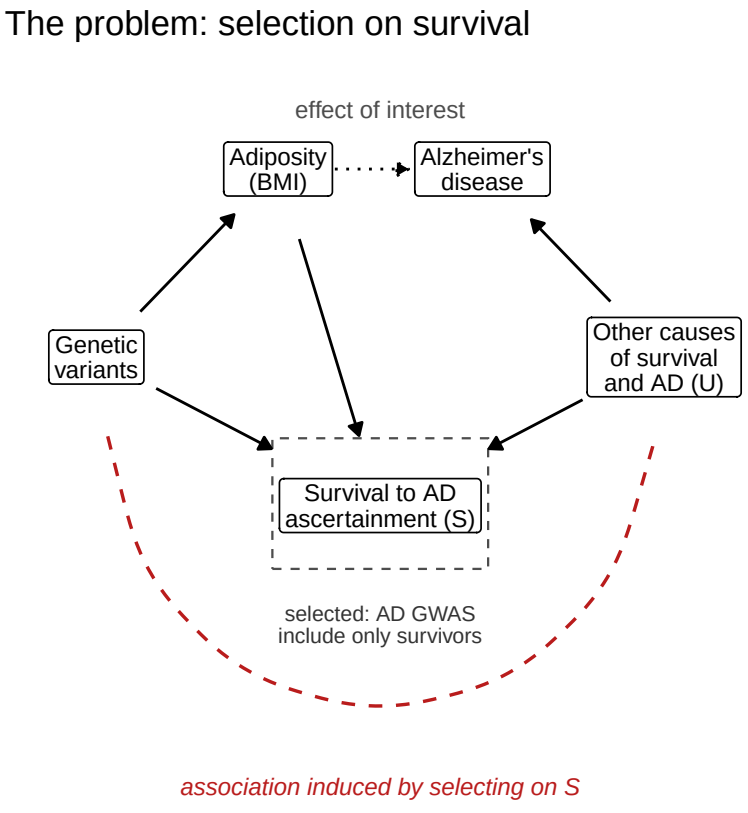
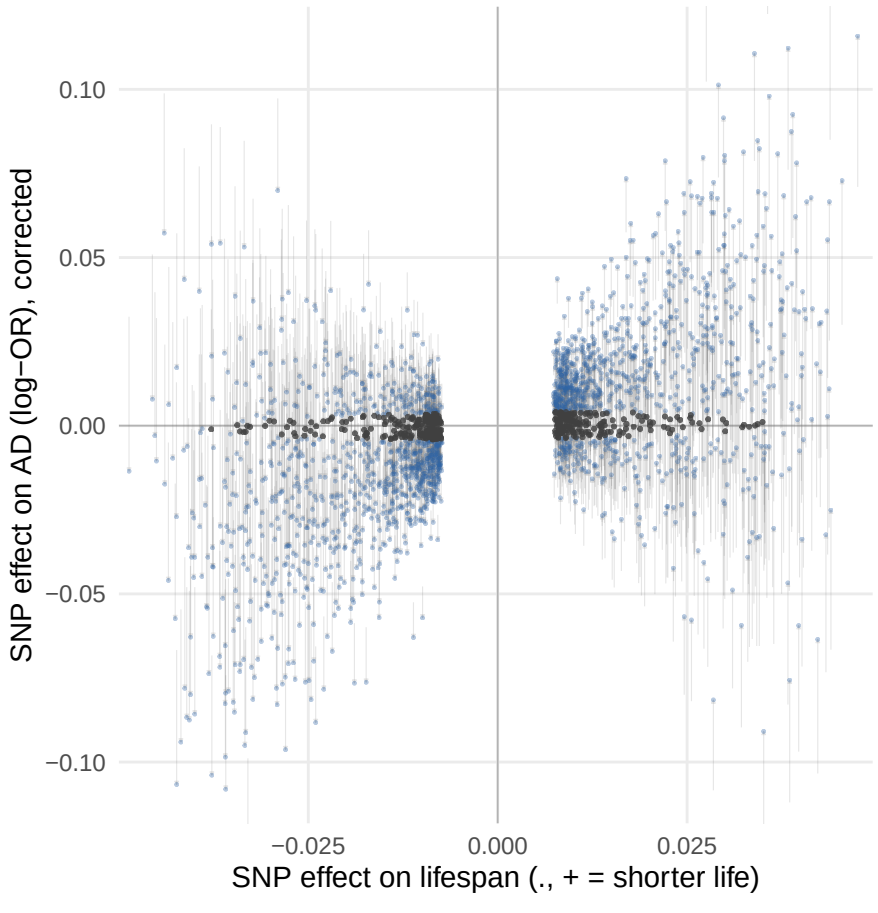
A


The problem: selection on survival
effect of interest
Adiposity (BMI)
Alzheimer's disease
Genetic variants
Other causes of survival and AD (U)
Survival to AD ascertainment (S)
selected: AD GWAS include only survivors
association induced by selecting on S

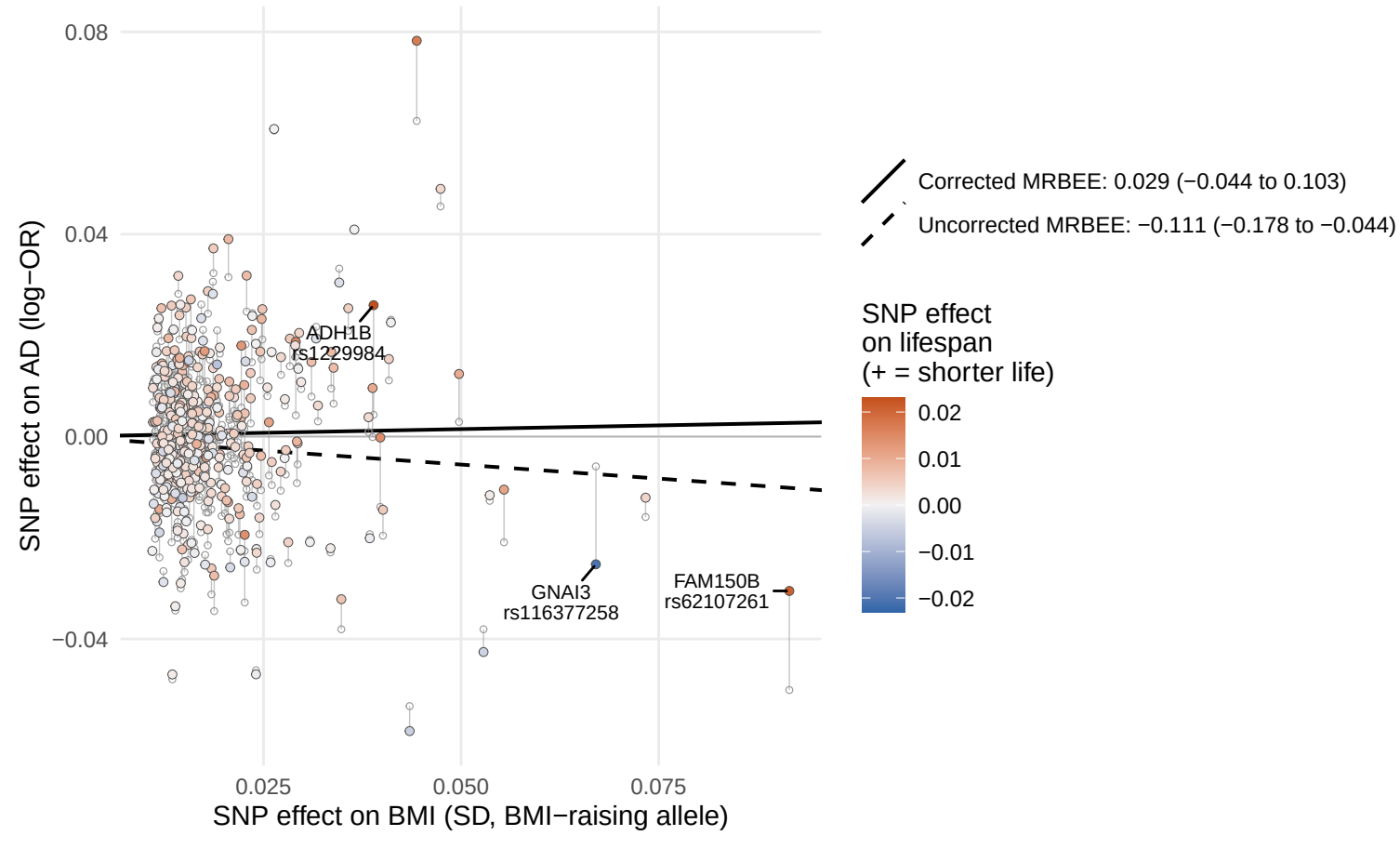
B


Estimating b: the slope of the survival−only cluster
estimated b = −0.94 (−1.08 to −0.81)
SNP effect on AD (log−OR), uncorrected
0.10
0.05
0.00
−0.05
−0.10
Pleiotropic (n = 1956)
Hunted: survival only (n = 370)
−0.025
0.000
0.025
SNP effect on lifespan (., + = shorter life)

C


Applying it: each SNP shifts by −b × its lifespan effect
SNP effect on AD (log−OR), corrected
0.10
0.05
0.00
−0.05
−0.10
−0.025
0.000
0.025
SNP effect on lifespan (., + = shorter life)

D


Consequence: BMI −> AD
SNP effect on AD (log−OR)
0.08
0.04
0.00
−0.04
ADH1B rs1229984
GNAI3 rs116377258
FAM150B rs62107261
0.025
0.050
0.075
SNP effect on BMI (SD, BMI−raising allele)
Corrected MRBEE: 0.029 (−0.044 to 0.103)
Uncorrected MRBEE: −0.111 (−0.178 to −0.044)
SNP effect on lifespan (+ = shorter life)
0.02
0.01
0.00
−0.01
−0.02